

# Age and systemic acid-base equilibrium: analysis of published data.

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To investigate whether systemic acid-base equilibrium changes with aging in normal adult humans, we reviewed published articles reporting the acid-base composition of arterial, arterialized venous, or capillary blood in age-identified healthy subjects. We extracted or calculated blood hydrogen ion concentration ( $[H^+]$ ), plasma bicarbonate concentration ( $[HCO_3^-]$ ), blood  $PCO_2$ , and age, and computed a total of 61 age-group means, distributed among eight 10-year intervals from age 20 to 100 years. Using linear regression analysis, we found that with increasing age, there is a significant increase in the steady-state blood  $[H^+]$  ( $p < .001$ ), and reduction in steady-state plasma  $[HCO_3^-]$  ( $p < .001$ ), indicative of a progressively worsening low-level metabolic acidosis. Blood  $PCO_2$  decreased with age ( $p < .05$ ), in keeping with the expected respiratory adaptation to metabolic acidosis. Such age-related increasing metabolic acidosis may reflect in part the normal decline of renal function with increasing age. The role of age-related metabolic acidosis in the pathogenesis of the degenerative diseases of aging warrants consideration.